

Cauchy point: minimizer of m_k along $-g_k$ (steepest descent)

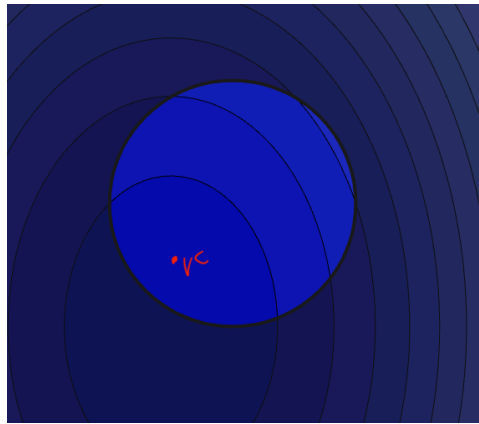
Problem 1:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with Cauchy point method.

curvature: $g^T B g = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 8 > 0 \Rightarrow$ pos. curvature. $\Rightarrow$ find analytic min inside Trust Region.

$$v^C = -t \Delta \frac{g}{\|g\|}, \quad t = \min \left\{ \frac{\|g\|^3}{\Delta g^T B g}, 1 \right\} \Rightarrow \|g\|^2 = \sqrt{2}^2 \approx 2.83 \Rightarrow t = \frac{2.82}{\frac{1}{2}(8)} \approx 0.71$$



$$\Rightarrow \boxed{v^C} = -0.71 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \boxed{\begin{pmatrix} -0.25 \\ -0.25 \end{pmatrix}}$$

$$\|v^C\| = t \Delta \approx 0.35 < 0.5$$

$\Rightarrow$ inside T.R. as expected

Problem 3:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Case 1: $\lambda_1 = 0$: KKT4) $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} v^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \Rightarrow \begin{cases} 4v_1^* = -1 \rightarrow v_1^* = -1/4 \\ 2v_2^* = -1 \rightarrow v_2^* = -1/2 \end{cases}$

$$\|v^*\|^2 = \frac{1}{8} + \frac{1}{4} = \frac{3}{8} < 1 = \Delta^2 \Rightarrow \text{so, } \boxed{v^* = \begin{pmatrix} -1/4 \\ -1/2 \end{pmatrix}, \lambda^* = 0}$$

$\Rightarrow$ within trust region, no need to consider $\lambda_1 > 0$.

KKT Conditions - using $\|v\|^2 \leq \Delta^2$

1.) $\|v^*\|^2 - 1 = 0$

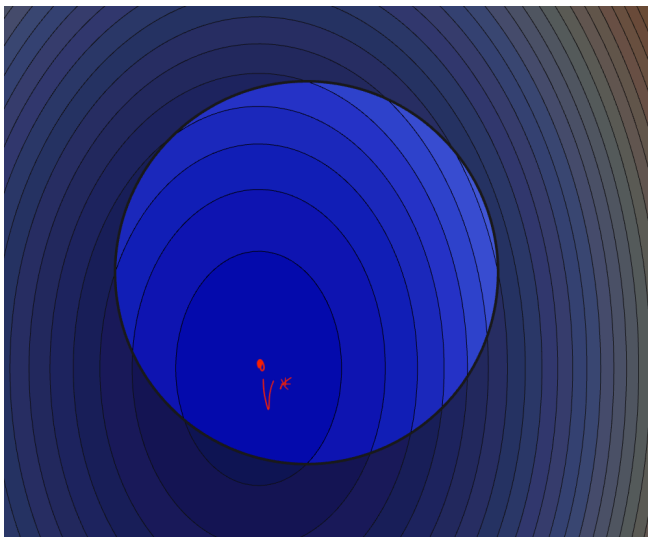
2.) $\lambda_1^* \geq 0$

3.) $\lambda_1^* (\|v^*\|^2 - 1) = 0$

4.) $\mathcal{L}(v, \lambda_1) = f + g^T v + \frac{1}{2} v^T B v + \lambda_1 (\|v\|^2 - 1)$

$$\nabla_v \mathcal{L}(v, \lambda_1) = g + Bv + \lambda_1 v \stackrel{!}{=} 0$$

$$\Rightarrow (B + \lambda_1 I) v^* = -g$$



KKT conditions: using $\|v\|^2 \leq \Delta^2$

Problem 5 (bonus):

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \Delta = 1/2$$

1.) $\|v^*\|^2 - \frac{1}{4} \leq 0$

2.) $\lambda_1^* \geq 0$

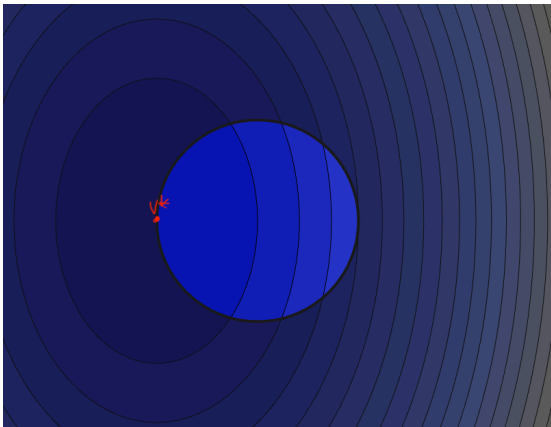
3.) $\lambda_1^* (\|v^*\|^2 - \frac{1}{4}) = 0$

4.) $(B + \lambda_1^* I) v^* = -g$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Case 1: $\lambda^* = 0$ KKT 4: $4v_1^* = -2 \Rightarrow v_1^* = -\frac{1}{2}$
 $2v_2^* = 0 \Rightarrow v_2^* = 0$ $\left\{ \begin{array}{l} \|v^*\|^2 = \frac{1}{4} = \Delta^2 \Rightarrow \text{on boundary} \end{array} \right.$

$\Rightarrow \lambda^* = 0, v^* = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}$



Problem 7: (bonus)

Same parameters as in problem 6, but now the gradient is zero $g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

KKT conditions:

1.) $\|v^*\|^2 - 1 \leq 0$

2.) $\lambda_1^* \geq 0$

3.) $\lambda_1^* (\|v^*\|^2 - 1) = 0$

4.) $(B + \lambda_1^* I) v^* = -g$

5.) $B + \lambda_1^* I \in S_+^n \Rightarrow \lambda_1^* \geq 2$

KKT 4.) $(4 + \lambda_1^*) v_1^* = 0 \quad (1)$

$(-2 + \lambda_1^*) v_2^* = 0 \quad (2)$

KKT 3.) From KKT 5 we know $\lambda_1^* \geq 2$, so $\|v^*\|^2 - 1 = 0 \Rightarrow v_1^{*2} + v_2^{*2} - 1 = 0 \quad (3)$

From (1), (2), (3): $\lambda_1^* = 2, v^* = \begin{pmatrix} 0 \\ \pm 1 \end{pmatrix}$ (using online solver)

